

2.3 Competitive Event: Permutation Sabotage Duel

- **The Concept:** When observing the training dynamics of the Perceptron Learning, it is expected to see convergence (there is even a theorem that guarantees that for linearly separable data), but the path taken and the number of mistakes depends on the **presentation order** of the training samples.
- **Red vs Blue (20 min):** Squads will compete for points and speed of calculation.
 - Each team need to choose between Red Team or Blue Team and inform the instructor BEFORE STARTING TO WORK.
 - Conclusion will only be considered if the teams present all calculation steps on paper.
 - Teams are not allowed to use computer programs (e.g. python, spreadsheets, etc) to speed up the calculations.
 - Simple calculators (sum/subtract/multiply/divide) are allowed.
 - RED Teams supply a permutation (ordering of indices $1, \dots, 6$) designed to **maximize** accuracy. Consider the first time it is measured higher than 80%, even if at the end of epoch 2 it is smaller.
 - BLUE Teams supply a permutation designed to **minimize** the total number of mistake updates in a 2-Epoch run (fastest convergence).
 - You are allowed to provide a different permutations for epoch 1 and 2, or just use the same pertumation on both epochs.
 - All teams execute their own proposed permutations starting from $\mathbf{w} = \mathbf{0}, b = 0$.
- Scoring
 - +50 AP per reduced mistake updates (normal sequence has 2).
 - +100 AP if your permutation reaches at least 80% accuracy.

You will work with the same calibration dataset $\mathcal{D}_2 = \{(\mathbf{x}^k, a^k)\}_{k=1}^6$:

- $\mathbf{x}^1 = [255, 128, 128, 0]^\top, \quad a^1 = 0$
- $\mathbf{x}^2 = [55, 128, 128, 128]^\top, \quad a^2 = 0$
- $\mathbf{x}^3 = [192, 128, 128, 0]^\top, \quad a^3 = 0$
- $\mathbf{x}^4 = [100, 128, 128, 100]^\top, \quad a^4 = 1$
- $\mathbf{x}^5 = [30, 64, 128, 30]^\top, \quad a^5 = 1$
- $\mathbf{x}^6 = [20, 64, 128, 0]^\top, \quad a^6 = 1$

Chosen color (RED / BLUE):

Chosen permutation:

1st epoch

Step k	Sample $\tilde{\mathbf{x}}^k$	Target a^k	Activation z^k	Output $\hat{a}^k$	Error e^k	Δb	$\Delta \mathbf{w}^\top$	New Bias b	New Weights $\mathbf{w}^\top$
1	$\tilde{\mathbf{x}}^1$	0							
2	$\tilde{\mathbf{x}}^2$	0							
3	$\tilde{\mathbf{x}}^3$	0							
4	$\tilde{\mathbf{x}}^4$	1							
5	$\tilde{\mathbf{x}}^5$	1							
6	$\tilde{\mathbf{x}}^6$	1							

2nd epoch

Step k	Sample $\tilde{\mathbf{x}}^k$	Target a^k	Activation z^k	Output $\hat{a}^k$	Error e^k	Δb	$\Delta \mathbf{w}^\top$	New Bias b	New Weights $\mathbf{w}^\top$
1	$\tilde{\mathbf{x}}^1$	0							
2	$\tilde{\mathbf{x}}^2$	0							
3	$\tilde{\mathbf{x}}^3$	0							
4	$\tilde{\mathbf{x}}^4$	1							
5	$\tilde{\mathbf{x}}^5$	1							
6	$\tilde{\mathbf{x}}^6$	1							